QC.1) Find Q_1 for $Q_2 = 0$ & $T_1 = 40^\circ C$:

$$\frac{dT_1}{dt} = \alpha_1 (T_\infty - T_1) + \alpha_2 \sigma (T_\infty^4 - T_1^4) + \alpha_3 (T_2 - T_1) + \alpha_4 (T_2^4 - T_1^4) + \alpha_5 Q_1$$

$\alpha_2 = \alpha_4 = 0$ & $\frac{dT_1}{dt} = 0$ gives

$$\alpha_1 (T_\infty - T_1) + \alpha_3 (T_2 - T_1) + \alpha_5 Q_1 = 0$$

The same for $\frac{dT_2}{dt}$: ($\alpha_4 Q_2 = 0$)

$$\alpha_1 (T_\infty - T_2) + \alpha_3 (T_1 - T_2) + \alpha_4 Q_2 = 0$$

$$T_2 = \frac{\alpha_1 T_\infty + \alpha_3 T_1}{\alpha_1 + \alpha_3} = \frac{0.003660 \cdot 18.5 + 0.001810 \cdot 40}{0.003660 + 0.001810}$$

$$= 25.614\ldots {}^\circ C \quad (4)$$

$$Q_1 = -\frac{\alpha_1 (T_\infty - T_1) + \alpha_3 (T_2 - T_1)}{\alpha_5}$$

$$Q_1 = 32.79\%$$